Erjona Topalli

Senior Full-Stack Software Engineer

React & TypeScript · Frontend Architecture · APIs & AWS · AI-Assisted Development

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Summary

Senior Software Engineer (promoted from Software Engineer) with 9+ years at Oracle building and modernizing large-scale, accessible web applications in React and TypeScript. Leads frontend architecture, API design, and AI-assisted development practices; mentors engineers and participates in hiring. MIT Computer Science & Mathematics graduate with a background in academic research (MIT CSAIL, Weizmann Institute) and competitive mathematics (IMO, BMO).

Experience

Oracle — Jul 2017 – Present

Senior Software Engineer (Senior MTS) — *2024 – Present*

- Delivered an end-to-end platform feature (Core Pack PictoChart) for Oracle JET, Oracle's open-source enterprise UI framework supported by a 5,500+ member internal developer community, spanning Preact, WebDriver testing, and documentation across 77 files – unblocking a downstream engineering workstream.
- Owned public API design for an interactive drill-down feature from architecture through release, clearing five cross-functional gates: architecture, performance, accessibility, testing, and documentation.
- Reduced shared visualization-layer coupling by migrating utilities across 5 chart components (Chart, Legend, Meter Bar/Circle, Rating Gauge, TagCloud), eliminating 7 redundant dependencies.
- Managed 47 versioned public API deprecations across 4 component families (Gantt, Chart/BoxPlot, NBox, Diagram), preserving backward compatibility for external consumers.
- Built Grafana observability dashboards to monitor application performance, enabling earlier detection and diagnosis of issues.
- Designed and built backend APIs and services outside the JET framework to support new product features, working across the full stack from service layer through UI.
- Mentored junior engineers through code reviews and architecture discussions; participated in engineering interview loops.
- Adopted AI-assisted development tools (Cursor, Codex) into daily workflow, authoring repo-level Cursor/Codex rules to standardize AI-assisted coding practices across the team.

Software Engineer (Member of Technical Staff) — *Jul 2017 – 2024*

- Designed and built responsive, accessible, high-performance web applications serving enterprise users across large-scale Oracle products.
- Built and owned 8+ reusable UI components and platform capabilities, improving developer productivity and long-term maintainability.
- Partnered with UX and accessibility specialists to design intuitive experiences meeting WCAG accessibility standards.
- Led proof-of-concept research that informed product direction and validated technical feasibility ahead of implementation.
- Promoted from entry-level to Member of Technical Staff within first year, then to Senior MTS in 2024 based on sustained technical impact and growing scope of ownership.

Oaktown Boulders — 2020 – 2022

Founding Engineer, Rock Climbing Gym Web App

- Built an IoT-integrated route-setting system for a climbing gym, connecting programmable LED-lit climbing holds to a backend over MQTT to enable real-time route design and lighting control.
- Implemented full CRUD functionality for creating, editing, and publishing climbing routes, allowing gym staff to set and load new routes directly through the app.
- Owned the end-to-end MEAN-stack architecture as sole/founding engineer, from database schema through UI, for a product used by gym members and staff.

Technical Skills

- **Languages:** TypeScript, JavaScript, Python, Java, HTML/CSS
- **Frontend:** React, Preact, d3.js, jQuery, Knockout
- **Backend & APIs:** Node.js, Express, MongoDB, AWS, API Design & Integration
- **AI & Tools:** Codex, Cursor, Figma, Git, AWS
- **Core Strengths:** Full-Stack Development, Frontend Architecture, Technical Leadership, Accessibility (WCAG), UI Performance Optimization, Data Visualization, Cross-Functional Collaboration

Education

Massachusetts Institute of Technology (MIT) — Class of 2017

BSc Computer Science & BSc Mathematics

Research & Honors

- **Undergraduate Research, MIT CSAIL** – "Visualizing Trackers for a User-Conscious Web," MIT SuperUROP (Hewlett Foundation Scholar), advised by Daniel Weitzner, MIT Internet Policy Research Initiative (2016–2017).
- **Undergraduate Research, MIT-Israel Program** – Advised by Prof. Moni Naor, Weizmann Institute of Science (2016).
- **International Mathematical Olympiad (IMO)** – Honourable Mention, representing Albania (2011).
- **Balkan Mathematical Olympiad (BMO)** – 2nd Place (2011), 3rd Place (2012), representing Albania.